

MSc Practicum / Research Practicum/ Internship Handbook

School of Computing

A Guide for Students and Supervisors

MSc in Artificial Intelligence

MSc in AI for Business

MSc in Cybersecurity (September 2023 & onward intakes)

MSc in Fintech (September 2023 & onward intakes)

MSc in Data Analytics (September 2024 & onward intakes)

MSc in Open Data Practice (January 2025 & onward intakes)

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Welcome from Dean of School

The National College of Ireland (NCI) has the reputation of being relevant to the needs of students and employers and hence developing excellence in the academic field.

The School of Computing has expanded greatly in the last few years, reacting quickly to changing needs and developments in the ICT sector.

Our Masters students come from nearly every continent and bring with them a vast array of experience. Some of you are coming straight from your undergraduate programmes; some of you will be making the giant leap of choosing to return to education after a long period outside of formal education. In short, the NCI postgraduate students are diverse and unique bringing with them a host of different experiences and life lessons that, when shared, create a fantastic learning environment.

Our teaching faculty demonstrate a blend of in-depth industry experience and, academic and research excellence thereby reflecting the needs of our students to gain access to cutting edge academic rigour while developing the necessary skills to apply these new insights to real world situations. Our Masters programmes are designed in such a way as to complement and advance the technical and research skill set of our students, to allow them to begin to shape their professional career trajectories in a meaningful way.

Postgraduate education is intense and challenging because it needs to be. There is much to learn in order to succeed in your chosen field of study. While the coming months will likely be filled with many long days, they should also be deeply satisfying. Studying as part of the School of Computing, you will learn about the latest technologies in dedicated computer labs. The School of Computing takes pride in the delivery of programmes using innovative problem-solving strategies to enhance the learning experience of our students. We also encourage our students to make the most of the non-academic elements of their study programme. Networking is an increasingly important part of any career development plan. Here at the College, we provide opportunities for our students to mix socially in order to develop and nurture those relationships during their time at the College and beyond graduation.

Ultimately NCI School of Computing is committed to providing an exceptional educational experience that will help you to achieve your goals. I hope that I will have the opportunity to meet you personally before I have the pleasure of shaking your hand at graduation. I wish you the very best in your studies at NCI and in your career beyond.

Professor Paul Stynes,

Dean of School of Computing

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1. Introduction

This handbook is intended to provide postgraduate students with a first stop resource to completing the Practicum/ Research Practicum/ Internship module (30 credits) in part fulfilment of a MSc Degree at Level 9 of the Irish National Framework of Qualifications¹.

Practicum module, Research Practicum module Internship module are part of a number of MSc courses as listed on the cover page of this document.

The aim of Practicum/ Research Practicum / Internship module is for learners to consolidate the knowledge and skills acquired in other modules by carrying out a capstone project that combines both research and technical skills; design, produce, and evaluate an innovative ICT solution related to a programme area.

The Practicum/ Research Practicum / Internship module (30 credits) consists of Part 1 and Part 2 and it is organised over two semesters for full time students, with workshops and assignment of supervisors taking place in Semester 2 (corresponding to Practicum/ Research Practicum / Internship Part 1), while Semester 3 consists exclusively of supervision (corresponding to Practicum/ Internship Part 2). Students who secured an internship role within a company will complete this activity during Semester 3 over a 3 month period. For part time students, Practicum/ Research Practicum / Internship Part 1 and Part 2 run over a 6 month period during Semester 4 of the programme.

To support those full-time students doing the Internship module in completing the internship portfolio, students have an additional 3 weeks after the 3 month internship within the company. This additional period does not apply for part-time students as they do the internship over six months.

In order to ensure the academic standards of all projects, every student will have an academic supervisor assigned for their *Practicum/ Research Practicum / Internship* project.

The Practicum/ Research Practicum/ Internship is intended to be a central and integrating element of the Masters programme at NCI. It is designed to afford postgraduate students the opportunity to exercise their creative, questioning, analytic, and writing skills as well as to develop research skills in a focused and practical manner. It is also intended to give students the chance to develop an in-depth understanding of some specific topics related to their chosen discipline and to apply research methods and techniques. One of the main features of Practicum/ Research Practicum/ Internship is that it must be beyond mere description.

People working in different roles within organisations are often required to write and deliver descriptive reports to their managers. This research-focused Practicum/ Research Practicum/ Internship project requires students to “move up a gear” from being a mere story or descriptive narrative to a critical analysis and evaluation of some topic. Typically, this “extra” dimension will require:

- One or more research questions in a particular domain to be identified
- The question(s) to have some theoretical backdrop
- The student to carry out both theoretical and empirical investigations

¹ See <http://www.nfq-qqi.com/>

- To propose, design, implement, evaluate, analyse, and discuss an ICT solution related to the programme domain that seeks out to find an answer to the identified research question(s)

The “question” element is initially addressed in the Practicum/ Internship Part 1 and it will be dealt with in greater details in the Practicum/ Research Practicum/ Internship Part 2.

As Practicum/ Research Practicum/ Internship is part of a postgraduate course it is also required that students employ and develop their research knowledge and skills in an applied fashion. The Practicum/ Research Practicum/ Internship must involve the identification, generation, or collation of relevant primary or secondary data and the ability to analyse them in a meaningful and critical manner.

1.1 Special Regulations

A student who selected Internship module and is unsuccessful at obtaining an internship with a company will undertake the Practicum module and will complete the capstone research project within NCI under the direction of the NCI academic supervisor.

If a student does not want to pursue the Practicum/ Internship module, the student can apply to exit the course with a Postgraduate Diploma award (Level 9) subject to the condition to pass all the taught modules (60 credits in total). The Exit Award Form is available in Appendix B. The form should be submitted to Registrar.

If a student would like to pursue the Internship module with a company/organization identified by the student, NCI approval is required prior to starting the Internship. Any interested student should provide the job description and discuss this option with the Programme Director and Careers Office before applying and engaging with the company. The internship work must be related to programme area and approved by the Programme Director in order to be officially considered as part of the Masters degree.

A research based project must be researched and developed as part of the Internship.

1.2 Practicum/ Research Practicum/ Internship Committee

The Practicum/ Research Practicum/ Internship Committee consists of the Practicum/ Research Practicum/ Internship module owner, MSc Programme Director, project supervisors, career office team and other academic members involved in the examination process.

The Practicum/ Research Practicum/ Internship Committee is responsible for managing the Practicum/ Research Practicum/ Internship process, with responsibility for screening and approving the Internship positions, monitoring progress, and ensuring completed internships are assessed and graded by a number of academics.

The chair of the Practicum/ Research Practicum/ Internship Committee (i.e., the Programme Director) is responsible for describing the operation of the Practicum/ Research Practicum/ Internship process to the students through the production of a handbook at the start of the academic year. This will involve describing the process to secure an internship, deliverables, due dates, and the grading rubrics.

The Vice Dean of Postgraduate Programmes & Research will identify appropriate academic members to supervise the capstone research project developed as part of the Practicum/ Research Practicum/ Internship Part 2 and will communicate to the students the assigned supervisors. The

academic supervisors will collaborate and will guide the students in researching, creating, implementing, testing and evaluation of the project.

Two examiners are also assigned to each student. One of the examiners will be the supervisor. The examiners independently fill in the research project assessment form and agree on a mark for the project. The supervisor compiles the comments of both examiners and fills the online marking form. The marking form and the agreed mark are reviewed and approved by Practicum/ Research Practicum/ Internship Committee.

2. Supervision

2.1 Appointment of Supervisor(s)

Students pursuing the Practicum/ Research Practicum/ Internship are assigned one academic supervisor. An industry supervisor will also be assigned by the company where the internship takes place for the Internship module. The academic supervisor assumes all the functions assigned to the 'supervisor' in these regulations.

The academic supervisor is a full-time academic member or an associate faculty member of the School of Computing, the holder of an award equivalent or higher than that being pursued, and he/she has supervised to the pursued award level.

In exceptional circumstances the Practicum/ Research Practicum/ Internship Committee may approve a supervisor not meeting these requirements.

The chair of the Practicum/ Research Practicum/ Internship Committee and the Vice Dean of Postgraduate Programmes & Research will identify appropriate academic members, will allocate supervisors, and will communicate to the students the assigned supervisor(s).

Students doing the Practicum/ Research Practicum/ Internship Part 2 will have to attend group supervision classes as per the timetables published in advance on the NCI's website.

2.2 The Supervisory Relationship

It is important to note that the academic supervisor assigned to a student will not necessarily be an expert in your chosen research domain. The supervisor will have sufficient experience to provide adequate guidance to a student on the production of their Practicum/ Research Practicum/ Internship project. A student however may approach any academic member or associate faculty within the School of Computing with a view to seeking out guidance on specific topics or issues pertaining to their research. However, it has to be understood that the first marker of an MSc project is the supervisor.

The supervisory relationship is an important one and it is essential therefore that good working relations are established and maintained at all times. There are several pointers that can help ensure that the relationship is a business-like and productive one for both parties. First, be sure to meet early in the process as soon as you have been assigned.

It is also important in the initial stages of the supervisory relationship that the supervisor and the student agree on the research topics to be investigated, a research plan and timescale, as well as the deadlines for submission of various milestones and deliverables part of the project.

Supervisors also like to read before a meeting any written material being brought by students. Therefore, from an early stage of your research work, make an attempt to always have some written material to bring along - written outline Practicum/ Research Practicum/ Internship project topics/ideas to be explored, written lists of articles you have read, a brief description of the tasks you have worked on since last meeting, a timetable for your work output, or individual sections – whatever is appropriate for the stage you are at.

Students should also avoid giving their supervisor rough drafts or “material I have to do more work on”. There is nothing more annoying for a supervisor than correcting some written material only to be told on handing it back: ‘oh I’ve reworked that completely since I gave it to you’. Supervisors can only make an informed and useful call on your work if what you submit is your best effort at that point in time.

2.3 What can Student expect?

In order to ensure as much clarity as possible around the roles of the student and the supervisor the following sections outline what are considered reasonable expectations of both parties.

The Practicum/ Research Practicum/ Internship project is the sole responsibility of each student. Students must engage in a proactive manner when undertaking their own research. It is the student’s responsibility to ‘drive’ their own research and take ultimate responsibility for the decisions they make in their research design and its implementation. The role of supervisor is similar to that of a consultant. They are assigned to offer students advice, counsel, and feedback regarding the progress of their Practicum/ Research Practicum/ Internship project.

2.3.1 Student Expectations of the Supervisor

Students can expect a supervisor to be committed to the Practicum/ Research Practicum/ Internship process and act in a professional manner in order to ensure high academic and ethical standards of work. The supervisor will be in a position to advise the student on the suitability of their chosen research topic and determine whether or not it has sufficient merit and scope to be researchable for the purposes of Practicum/ Research Practicum/ Internship project in relation to the award of a Masters degree. Supervisor will primarily be accessible to students during the timetabled supervision classes, and students also need to understand that faculty members may not be available outside this time and any additional meetings need to be scheduled and agreed in advance. It is not acceptable to just knock at the supervisor’s office door every time you encounter difficulties with your research work and demand the supervisor to talk to you about the project.

At the start of the supervision process, the supervisor needs to discuss and agree with the student their mutual expectations. The supervisor and student need to agree a program of work and an initial timeframe of how often, when, and how communication is to occur. It is advisable that the supervisor and student agree the ground rules (i.e., explain their mutual expectations to each other). They should also agree on a schedule of deadlines and a series of milestones and deliverables as part of the project. The School follows a group supervision model, where typically meetings between supervisors and students occur in timetabled slots and deadlines for deliverables are decided in advance by the Practicum/ Research Practicum/ Internship Committee.

Also, academic members of School may take leave during the period of supervision, in which case timetabled supervision classes may be rescheduled to another date. Supervisors are not required to

be available to students at all times and hence it is prudent to ask your supervisor if, and when, they intend taking annual leave so that the student can plan adequately for this.

It is not the responsibility of the supervisor to ensure that their student completes and submits their Practicum/ Research Practicum/ Internship project and the required documentation. This is the sole responsibility of the student. A supervisor is there to offer guidance, support and to act as a sound board.

Students are solely responsible ultimately for the quality of the work they produce and the completion of their Practicum/ Research Practicum/ Internship project portfolio.

2.3.2 Supervisor Expectations of the Student

In part fulfilment of the requirements for the award of a Masters, students are expected to undertake independent research and be responsible for their own research activity. This responsibility is the student's and students alone.

Students are expected to attend all timetabled supervision classes, maintain regular contact with their supervisor, reply to any type of communication received from the supervisor, agree, and keep deadlines and report regularly on their progress.

Full time students are expected to fully dedicate their time for completing their Practicum/ Research Practicum/ Internship project work and no holidays should be taken.

Students are also required to document minutes of the meetings that take place with their supervisor. This documentation will ensure that the student understands the feedback given to them each session.

Students also need to be aware of academic regulations and guidelines in relation to the submission of their Practicum/ Research Practicum/ Internship project. It is not the responsibility of the supervisor to ensure that a student adheres to these however they may offer guidance on what these regulations are and where further information can be found.

Students are required to attend and to actively participate in all classes scheduled for taught module(s) that will run in the same semester(s) as the Practicum/ Research Practicum/ Internship module.

2.3.3 Supervision Grievance Procedures

A student should make a complaint about the adequacy of supervision in the first instance to their supervisor. Often problems arise as both parties have different expectations of what is appropriate and required. A discussion around expectations and the issues experienced can often be enough to resolve the problem.

Where a problem persists, or the parties cannot agree then the matter should be referred to the Programme Director followed by the Vice Dean of Postgraduate Programmes & Research. Please note, however, that the College is not in a position to offer students a choice of supervisor and hence only serious issues of an academic nature will be tolerated. Complaints about the adequacy of supervision will not be entertained once the Practicum/ Research Practicum/ Internship project has been presented for examination i.e., post submission.

3. Practicum/ Research Practicum/ Internship Objectives

The aim of the Practicum/ Research Practicum/ Internship module is for learners to consolidate the knowledge and skills acquired in other modules by carrying out a project that combines both research and technical skills within a workplace; to develop the practical competencies and communication skills necessary to function as a team member in a workplace. The learners will investigate, design, produce and evaluate an ICT solution related to the programme area.

It is also expected that knowledge and skills gained will contribute not only to scholarship, but also to the generation of innovative solutions to intuitive research questions.

Specifically, the *learning outcomes* are:

- Propose a research question, project objectives and methodology.
- Analyse, synthesise, and critically evaluate the state of the art.
- Propose, architect, implement, and evaluate an ICT solution related to the programme area.
- Investigate potential future research and invention disclosures.
- Present and defend the project findings through a viva, artefact/product demo, and report.
- Understand the ethical issues that need to be addressed when conducting research.
- Demonstrate initiative whilst working alone or part of a team, and appropriate communication and interpersonal skills.

The Practicum/ Research Practicum/ Internship module (30 credits) consists of Part 1 and Part 2 and it is organised over two semesters for full time students, with workshops and assignment of supervisors taking place in Semester 2 (corresponding to Practicum Part 1/ Internship Part 1), while Semester 3 consists exclusively of supervision (corresponding to Practicum Part 2/ Internship Part 2). For part time students, Practicum/ Research Practicum/ Internship Part 1 and Part 2 run over a 6 month period during Semester 4 of the programme.

The student has the responsibility to participate in all scheduled meetings organised as part of Practicum/ Research Practicum/ Internship module.

Practicum module, Research Practicum module and Internship module have the same assessment strategy and requirements.

Students have to *submit a portfolio* (as part of the module assessment) that consists of:

Practicum Part 1/Research Practicum Part 1/ Internship Part 1:

- A project proposal with ethics approval
- Ethics Form(s)
- Cloud Resources Requirements Form (if NCI's public cloud resources will be used)

Practicum Part 2/Research Practicum Part 2/ Internship Part 2:

- A research paper style report
- Weekly activity reports
- An artefact / product / software developed
- A configuration manual
- Viva video(s) recording that consists of presentation video & artifact demo video
- Student answers to the questions given by the examiners

The weekly report (see Appendix H) must demonstrate some work progress was achieved on the project during a particular week. Student must show the weekly report with a summary of activities

performed at the beginning of the supervision session. Student must submit on Moodle the weekly progress report at the end of each supervision session, including the supervisor feedback.

Students who do the Internship within a company also have to submit weekly reports that describe student's work activities in the designated business environment and project progress.

More details on the required assessment components will be presented in the Assessment section.

3.1 Internship Process

All students are enrolled into Practicum/ Research Practicum module by default. All students are required to attend the class sessions and submit the assessments part of Practicum Part 1 / Research Practicum Part 1 as scheduled in their timetable.

Students have to express their interest to do Internship instead of Practicum and to participate in the workshop activities related to Internship, organised by NCI Careers Development and Employability Office. Students who successfully secure an internship job approved by the College will be transferred from Practicum/ Research Practicum into Internship and they will complete Internship Part 2 under an academic supervisor.

If a student wants to secure their own Internship, the job must be approved by Programme Director well in advance of the semester when Internship Part 2 is due to start. Please check the Programme Moodle page for submission deadline of the required documentation to have the Internship position approved. For full-time students the duration of the internship job is one semester (as part of Internship Part 2) during which time students are required to work full-time within the company 4 days each week (Monday to Thursday) and attend the 5 credits taught module classes as well as MSc project supervision classes 1 day (on Friday) each week. After the semester has finished, students are given 3 additional weeks to finalise the project portfolio to be submitted.

For part-time students the duration of the Internship is six months, and the supervision classes may be scheduled in the evening or weekends.

An academic supervisor will be assigned as the main supervisor. The industry supervisor assigned by the company will also be part of the supervisory team.

The MSc research project work shall build upon the practical work conducted in the company as part of the internship and extend this with the required research elements (e.g., research question, literature review, evaluation, etc.).

The student should engage as soon as possible with the industry supervisor and obtain details on their specific work they will have to do within the company. Moreover, students must attend the supervision classes in NCI and discuss with the academic supervisor on how to expand on that work and add research elements so that it has sufficient merit and scope to be researchable for the purposes of a research project in relation to the award of an MSc degree.

Students who chose Internship but were not successful in securing a job by the start of the semester when Practicum Part 2/ Research Practicum Part 2/ Internship Part 2 runs, will be required to do Practicum Part 2/ Research Practicum Part 2

3.2 Practicum/ Research Practicum

The Practicum/ Research Practicum module (30 credits) consists of Part 1 and Part 2 and it is organised over two semesters for full time students, and over a 6 month period during Semester 4 for part time students.

All students are enrolled into Practicum/ Research Practicum module by default. All students are required to attend the class sessions and submit the assessments part of Practicum Part 1 / Research Practicum Part 1 as scheduled in their timetable.

Practicum Part 1 / Research Practicum Part 1 runs during the entire semester 2 for full time students and as block delivery in Semester 4 for part-time students.

The duration of the Practicum Part 2 is one semester for full time students. For part-time students the duration of the Practicum Part 1 and 2 is six months.

As part of Practicum Part 2/ Research Practicum Part 2, students will work on research project under the direction of the NCI academic supervisor only.

The research domain and topic are defined and shaped as part of the research activities required for the Practicum/ Research Practicum Part 1. It is student's responsibility to identify and propose the research topic of the research project. The lecturer who delivers Practicum Part 1/ Research Practicum Part 1 will provide guidelines and advice to the student on the suitability of their chosen research topic and determine whether it has sufficient merit and scope to be researchable for the purposes of a research project.

Based on the submitted research proposal as part of the Practicum Part 1/ Research Practicum Part 1, an academic supervisor(s) will be assigned. The student has the responsibility to participate in all timetabled supervision classes.

4. Managing the Practicum/ Research Practicum/ Internship

This section provides students and supervisors with a framework in which to plan, write and complete the Practicum/ Research Practicum/ Internship.

4.1 Securing the Internship

The NCI Careers Development and Employability Office has a team of senior recruitment experts and guidance professionals that are experienced in providing tactical, customised, and comprehensive career development and personal branding advice and guidance to mid-senior level professionals in areas including career acceleration, career change and career crisis.

NCI is proud to achieve "Highly Commended" in the AHECS Excellence in Employability Awards 2020. NCI Career Development & Employability Office won the AHECS "Excellence in Employability Award" for 2013, 2014, 2016, and 2018.

Each programme that incorporates an internship module is assigned a dedicated internship advisor who is responsible for monitoring the students, conducting one to one career coaching sessions, CV preparation, interview preparation techniques and advice on job and internship suitability.

NCI Career Development and Employability office also provide extra support by organising on-site Career Fairs, Industry speakers and Life Coach Speakers; by arranging industry specific employability events such as a Whiteboard Interview Seminar where students get first-hand experience on the structure and procedure around whiteboard interviewing techniques.

The Career Development and Employability Office is the students' partner in establishing crucial relationships with employers and alumni and the majority of our students credit the NCI Careers Service as their primary job source.

All learners enrolled into the Masters programmes that have an Internship module are encouraged to engage in the internship process from the beginning of the programme. Internship positions are obtained by the Careers and Employability Office and are generally carried out over a period of three months in full-time mode.

The internship process consists of the following:

- Students submit final CVs to the internship advisor the semester before the internship commence.
- Employers advertise internship opportunities with the internship advisor.
- The internship advisor posts placement opportunities on the NCI portal.
- Students are expected to check the NCI student portal on a regular basis and to apply to the opportunities advertised by emailing a tailored CV to the internship advisor.
- The internship advisor will forward CVs to the employer.
- The employer will select the students they wish to interview.
- The internship advisor will contact the students that have been selected for interview and co-ordinate these interviews.
- The student will contact the internship advisor with interview feedback immediately after the interview.
- The internship advisor will liaise with the employer post interview, collate feedback, and deliver offers to successful students.
- Students are required to attend all interviews arranged for them by the internship advisor. Failure to attend for interview may result in the student being eliminated from the internship process and subjected to disciplinary proceedings within NCI.

Getting an internship is a competitive process, thus students are encouraged to actively search and try to secure their own internship with a company. However, any student that identifies an internship position should e-mail the internship job description and completed secure own internship form (see Appendix G) to the career advisor and Programme Director and discuss this option with the Careers Office before applying and engaging with the company.

If the internship job description is approved by NCI and the student is successful in obtaining the internship, the agreement of the company/organisation to participate in the Internship programme is required prior to starting to work on the internship.

Part-time students that are working in the IT sector can undertake the Internship within the company they work for by being involved in a Masters programme domain related project. However, the Internship role must be approved by Programme Director in advance. Please check the Programme Moodle page for further details including deadlines for submission of the Internship related documentation.

4.2 Internship Contract

A contract between learner, College and where applicable the employer describes the duties of each party (see Appendix E). Among other things, learners are required to bring any issues or barriers encountered to the attention of internship coordinator so that the College can intervene in time to make sure the learner is given the opportunity to complete their internship.

Note: If the student leaves his Internship position without NCI approval, no NCI support will be provided to secure another job. The student will also fail the Internship module.

4.3 Planning the Research Activities

The key to completing the Practicum/ Research Practicum/ Internship project on time and without compromising quality is to carefully manage your time and undertake the research in a rigorous and professional manner. Planning your research involves identifying the tasks that need to be undertaken as part of the research and planning where and when to work on particular aspects of the project. The Practicum/ Research Practicum/ Internship Part 1 class sessions teach the students research related concepts and develop skills on how to approach the following tasks: define a research question, create a literature review, research and development design and methodologies, and make use of statistical tools to critically assess research outputs and results. Therefore, students are required to use in the Practicum/ Research Practicum/ Internship Part 2 the knowledge acquired in Practicum/ Research Practicum/ Internship Part 1.

As the project progresses toward the final stage, it is likely that it will become more focused on writing tasks. Starting to write at an early stage on the research report ensures that you get used to the routines of writing. This can help you to complete your Practicum/ Research Practicum/ Internship project in a timelier manner.

It is also important to set realistic and achievable deadlines when working on the project and to be willing to revise them as the research progresses.

The Practicum/ Research Practicum/ Internship project involves a high level of independent study and learning with a significant number of hours dedicated for the project work. By delaying getting started to work on the project or making excuses as reasons for not starting the work, the quality of the project will be compromised, and deadlines may be missed.

The challenge, particularly for part-time students who are attempting to complete a research project while dealing with the day to day demands of daily life, is to find the time to dedicate to their study.

The key to success is setting aside adequate time on a weekly basis and being disciplined in terms of sticking to these commitments.

The use of GitHub is encouraged as a collaborative tool to enable students to work on the development of the project. It will also improve the transparency of projects for examiners as it will enable them to clearly identify the work carried out by students as part of the project.

4.4 Research Ethics

Any research involving human participants or/and secondary dataset(s) that is conducted by students at the National College of Ireland should be done so in an ethical manner. The college has

therefore developed an Ethics Committee, which acts as a sub-committee of the Research Committee, to ensure that ethical principles pertaining to research involving human participants are upheld and adhered to. School of Computing Ethics Filter Committee is a school specific group part of the Ethics Committee and is responsible with management of the ethical aspects specific to student projects.

4.4.1 Research Involving Human Participants

All students intending to use human participants as part of their projects must consult the College's Research Ethics Committee if the research project involves the following:

- direct experimentation on individuals.
- surveys or questionnaires administered to individuals.
- use of data derived from individual records where individuals might be identified.
- experimentation on animals.

Human participants may be directly involved in various stages of your project (e.g., requirements specification, design, development, evaluation, etc.) by taking part in interviews or brainstorm sessions or by answering surveys.

Ethical clearance must be received before the interaction with human participants commences.

As part of Practicum/ Research Practicum/ Internship Part 1 students must submit the following two documents (see Figure 1) on the Practicum/ Research Practicum/ Internship Moodle page that will be reviewed by the assigned lecturer.

- Declaration of Ethics Consideration Form (Appendix C)
- Ethics Application Form (Appendix D)

The submission deadline is communicated by the lecturer on the Research Project Moodle page. Template documents for these forms as well as examples of completed forms are provided on the Research Project Moodle page.

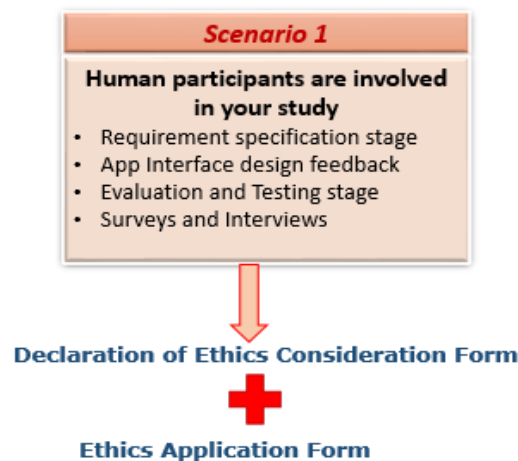


Figure 1 Forms required to be submitted when research involves direct interaction with human participants

Following consideration of research project submitted for ethical approval, a decision is made regarding the ethical risk level. For research that is deemed to fall under low ethical risk (Research Category A), a favourable outcome given by the supervisor will be sufficient to secure ethical approval. Research falling under the other two categories (Research Category B and C) must however be considered by the School of Computing Ethics Filter Committee and by the Ethics Committee before approval may be granted. The supervisor will inform the School of Computing Ethics Filter Committee about the student's project high ethical risk level.

More details regarding Research Ethics and the procedures required for seeking ethical approval of research involving human participants can be found on the NCI student portal in the [Ethical Guidelines and Policies for Research Involving Human Participants.pdf](#) document available under the Academic Policies web page.

4.4.2 Research that Makes use of Secondary Dataset(s)

If the research project does not involve human participants but public or private secondary dataset(s) are used in the research, Declaration of Ethics Consideration Form (Appendix C) must be submitted. The form requires the student to demonstrate that he/she has permission to access, to process and to analyse the dataset(s) and to make public the research findings.

Ethical clearance must be received before starting to process and analyse the dataset(s).

The dataset(s) could be provided by an individual that owns the dataset; could be downloaded from a web site or could be provided by a company. If the student generates a mock dataset, ethical clearance is not required as the student is the owner of the dataset.

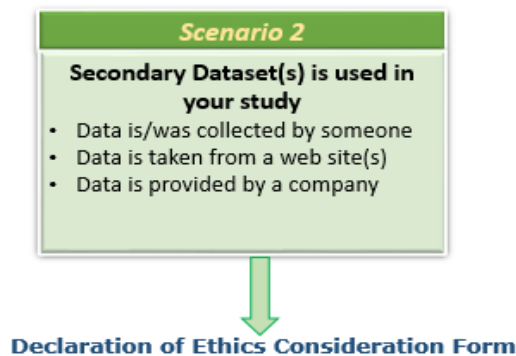


Figure 2 Forms required to be submitted when research involves using public and/or private datasets

Students must submit the following document (see Figure 2) on the Research Project Moodle page that will be reviewed by the assigned supervisor.

- Declaration of Ethics Consideration Form (Appendix C)

The submission deadline is communicated by the relevant MSc Programme Director on the Research Project Moodle page. Template documents for this form as well an example of a completed form is provided on the Research Project Moodle page.

Following the assessment of the Declaration of Ethics Consideration Form the ethical clearance may be granted or extra evidence may be required. If the form does not demonstrate that the dataset(s) can be used by the student for research purpose ethical approval is not granted. In this situation, the student must bring new evidence or change the source of the dataset(s).

4.4.3 Research that Makes use of Target Systems(s)

A research project that makes use of public or private real target system(s) does not need ethics permission but needs a letter/email from the target system owner regarding potential use. Declaration of Ethics Consideration Form (Appendix C) must be submitted. The form requires the student to demonstrate that permission/authorization has been granted to the student for using the system. In addition to this, in the cases where the goal of the project is to conduct penetration testing, vulnerability scans, or other ethical hacking engagement, the student must also provide a document describing the rules of engagement.

Ethical clearance must be received before starting to use the target system(s).

Target system refers to any asset, device, infrastructure component, hosting environment, network, or application, that is intended to be used by the current researcher as the target system for testing their ICT solution artefact. This can include a penetration testing target or any system under test.

Rules of Engagement is a document that contains the formal approvals, authorizations, scope, and other general guidelines or formal objectives necessary to execute a penetration testing engagement.

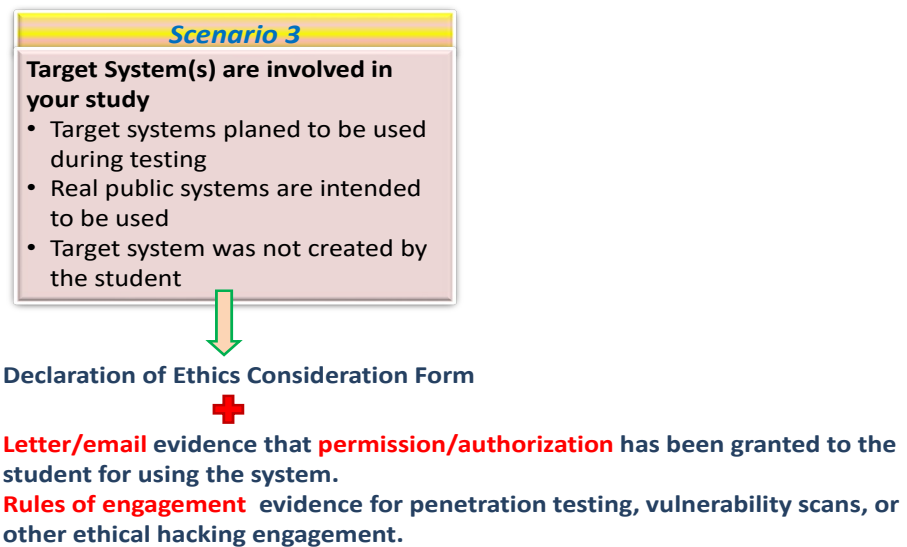


Figure 1 Forms required to be submitted when research involves target systems or hacking.

Students must submit the following document (see Figure 3) on the Research Project Moodle page that will be reviewed by the assigned supervisor.

- Declaration of Ethics Consideration Form (Appendix C)

The submission deadline is communicated by the relevant MSc Programme Director on the Research Project Moodle page. Template documents for this form as well as an example of a completed form is provided on the Research Project Moodle page.

Following the assessment of the Declaration of Ethics Consideration Form the ethical clearance may be granted or extra evidence may be required. If the form does not demonstrate that the target system can be used by the student for research purpose ethical approval is not granted. In this situation, the student must bring new evidence or change the target system and/or Rules of Engagement

4.5 Problems with Academic Writing

Some students may encounter difficulties when they have to write an academic report. Most common problems faced while writing a document are difficulty in formulating what to write, difficulty with bringing ideas into a coherent shape, not having the academic writing skills, etc. Issues of this nature need to be addressed at an early stage to prevent them from developing into serious difficulties.

Here are some practicable solutions that can be tried.

- Perform writing tasks at an early stage of your research project in order to increase confidence and writing skills. Examples of writing tasks are reviews of literature create notes on your research findings/ test results, write brief analysis of the read papers/articles.
- Request feedback from your supervisor on what you have written. Supervisors can help by given guidelines on writing tasks, including the length, scope, content, structure and aims of each task.
- Meet and discuss with your colleagues the research findings, share writing experiences or discuss and critique an article.

Students may also look for support from the NCI Learning Development Service part of the NCI's Student Services and Support Department. More details can be found on NCI student portal and NCI website

(<https://www.ncirl.ie/learningandteaching/academicsupport>).

5. Assessment of the Practicum/ Research Practicum/ Internship

The Practicum/ Research Practicum/Internship module consists of Part 1 and Part 2 and it is organised over two semesters (Semester 2 and Semester 3) for full time students. For part time students, Practicum/Research Practicum/ Internship Part 1 and Part 2 run over a 6 month period during Semester 4.

Practicum module, Research Practicum module and Internship module have the same assessment strategy and requirements.

The following documents are required to be submitted electronically online on the module Moodle page, as part of the Practicum/ Research Practicum/Internship module assessment, no later than the provided submission deadline.

Practicum Part 1/ Research Practicum Part 1/ Internship Part 1:

- A project proposal with ethics approval
- Ethics Form(s)
- Cloud Resources Requirements Form (if NCI's public cloud resources will be used)

Practicum Part 2/Research Practicum Part 2/ Internship Part 2:

- A research paper style report
- Weekly activity reports
- An artefact / product / software developed
- A configuration manual
- Viva video(s) recording that consists of presentation video & artifact demo video
- Student answers to the questions given by the examiners

The assessment documents submitted as part of Practicum Part 1/ Research Practicum Part 1/ Internship Part 1 will be graded by the lecturer responsible to deliver the lecture and tutorial sessions. Two examiners will review the documents, and the videos submitted as part of Practicum Part 2/ Research Practicum Part 2/ Internship Part 2. The student will be provided with a set of questions from the two examiners in relation to their work. The students must upload their answers on Moodle before the indicated deadline.

Important Note: The College does reserve the right to request a live viva session to take place in the event where examiners feel that further clarification is required regarding a student submission.

Therefore, **ALL** students will be required to make themselves available for a possible live viva session that would take place at date specified well in advance on the Moodle page.

Weekly activity reports: Students must provide a summary of weekly research activities performed on the project in the supervision meetings. Reports must be submitted on Moodle page, on a weekly basis.

5.1 Marking Schema

The Practicum/ Research Practicum/ Internship module marks are determined using the following schema.

MSc programme: Full time delivery mode				
Assessment Type	Assessment Description	% of total	Assessment Date	
Practicum Part 1/ Research Practicum Part 1/ Internship Part 1	Project Proposal and Ethics Approval	10%	Semester 2	
Practicum Part 2/ Research Practicum Part 2/ Internship Part 2	Project Specification	5%	Semester 3	
	Literature Review	10%	Semester 3	
	Artefact/Product Development	30%	Semester 3	
	Artefact/Product Evaluation and Analysis	25%	Semester 3	
	Document Presentation/Structure, Referencing, and Configuration Manual	10%	Semester 3	
	Viva	10%	Semester 3	

MSc programme: Part time delivery mode				
Assessment Type	Assessment Description	% of total	Assessment Date	
Practicum Part 1/ Research Practicum Part 1/ Internship Part 1	Project Proposal and Ethics Approval	10%	Semester 4	
Practicum Part 2/ Research Practicum Part 2/ Internship Part 2	Project Specification	5%	Semester 4	
	Literature Review	10%	Semester 4	
	Artefact/Product Development	30%	Semester 4	
	Artefact/Product Evaluation and Analysis	25%	Semester 4	
	Document Presentation/Structure, Referencing, and Configuration Manual	10%	Semester 4	
	Viva	10%	Semester 4	

5.2 Project Proposal and Ethics Approval

As part of Practicum Part 1/ Research Practicum Part 1/ Internship Part 1, learners submit a portfolio that consists of

- Project Proposal.
- Ethics Form(s)
- Cloud Resources Requirements Form (if NCI's public cloud resources will be used)

A project proposal must describe what the learner will investigate, the importance and relevance of the topic to the programme area, the research question(s) and how the learner intends to fulfil the project goals. The format of a project proposal varies between fields, but most proposals should contain at least these elements: introduction, research question, a brief literature review, and high-level description of the proposed research methodology and evaluation methodology.

The proposal has to address the ethics of the proposed study and the proposal has to have ethics approval. Students are required to fill in and submit via Moodle an Ethics Declaration form(s) regardless of if they use or not public/private secondary data set or/and involve human subjects and/or target systems. There are specific templates available on Moodle that you must use for this submission. More details about Research Ethics requirements are provided in Section 4.4

Students who identify that they will need NCI's public cloud resources for their research project must submit a Cloud Resources Requirements Form so that IT can make them available in a timely manner. There is a specific form available on module's Moodle that you must fill in.

5.3 Practicum/ Research Practicum/ Internship Weekly Activity Report

As part of the supervision process, students must submit weekly activity report that demonstrate some work progress was achieved on the project during a particular week. Student must show the weekly activity report with a summary of activities performed at the beginning of the supervision session. Student must submit on MSc Practicum/ Research Practicum/ Internship Moodle page the weekly progress report at the end of each supervision session, including the supervisor feedback.

Students who do an Internship within a company are also required to submit weekly activity reports that describe student's work activities in the designated business environment and project progress. A template of the weekly activity report is provided to the students on the Moodle page. (Appendix H)

5.4 Research Paper Style Report

A research paper style report describes the individual research project specification and software/artefact development as well as the testing/evaluation results. A literature review should also be included to ensure the validity of the research area. Students are also required to critically analyse insights gained throughout evaluation of their software/artefact development.

The research paper style report shall comprise up to 20 pages (approx. 4,000 to 6,000 words) and describe the individual research outcome and production of an ICT solution. This report shall build upon the work/tasks conducted in the company or the work conducted as part of Practicum/ Research Practicum/ Internship Part 1 updated and revised based on feedback from supervisor(s).

Note that direct copy-paste or paraphrasing from the Practicum/ Research Practicum/ Internship Part 1 proposal is not permitted as you already received marks and credits for that work. Failure to update and expand substantially the sections that overlap with the research proposal will result in penalties and mark deductions.

The students may be penalised due to superfluous text and are strongly advised to stay within 10% of the given page limits.

It shall follow the structure of a research paper that includes abstract, introduction, literature review, research methodology, design and implementation, evaluation, conclusion, and discussion. The report template provided on the module's Moodle page must be used when editing the research paper style report.

5.4.1 Report Structure

It is expected that the report to be submitted to consist of the following key components:

- *Abstract:* The role of the abstract is to summarise the project you have undertaken in a concise manner, highlighting the scope of the project, the objectives, the methodology and its finding and results.
- *Introduction:* the main purpose of the introduction is to motivate the contribution of the research work. It should include background on the research topic, the justification for the research and what gap in the literature it seeks to fill, the main research questions and hypotheses, and the organization of the study.
- Up-to-date *Review of the Literature/Background/ Literature Survey:* The literature review should demonstrate evidence of independent research critically analysing the potential of an application / idea and provide insights into how it can be implemented and evaluated. This component shall build upon the literature review work conducted as part of Practicum/ Internship Part 1 and shall be updated and revised based on feedback from supervisors and based on the final developed product.
- *Research Methodology:* The main purpose is to outlines the proposed research approach and method, justifies the choice of method(s) leveraging appropriate literature to substantiate choices / assumptions and/or key decisions, and considers the limitations and strengths of that approach, as well as demonstrating how the method will be applied to answering the research question(s).
- *Design Specification:* The techniques and/or architecture/framework that underlie the implementation and the associated requirements are identified.
- *Implementation/Solution Development:* The purpose is to describe only the final stage of the implementation. It must not contain code listings or a user manual/guide.
- *Evaluation:* A comprehensive analysis of the results and main findings of the study using multiple strategies as well as the implications of these finding both from academic and practitioner perspective are presented. For example, an algorithm may be comprehensively benchmarked using novel performance specific metrics on the Cloud Competency Centre facilities or external cloud providers; an internet application or mobile application may be evaluated using suitable usability testing techniques. Statistical tools should be used to critically evaluate and assess the experimental research outputs and levels of significance.

Due to page number limitation only the most relevant results are analysed and presented as evidence to support or reject the hypothesis.

- Conclusions and Discussion: A brief reiteration of the main research questions addressed and what was found as outcome of the research work is provided. Insights gained throughout the development and evaluation of the proposed solution are critically analysed and discussed. In addition, further avenues for research and commercialization should be identified.
- References: Correct citation references must be included
- Appendix: Configuration manual should be provided as a different file from the research paper style report, and it will not count towards the suggested word count.

Remember that all figures, tables, photographs etc. should be referenced and numbered in the report e.g., the graph (Fig 1) which complements the Table (Table 1).

It is highly recommended that students take some time at the start of the research process to go to the library and review other MSc research projects. This will give you a feel for the type of appropriate research questions, the research approaches adopted and the format of the report.

The research report should NOT contain a table of contents. If appropriate, the configuration manual may contain a table of contents.

5.4.2 Literature Review

The purpose of the literature review is to convey to the reader what knowledge and ideas have been established on a set of research topics from a particular domain, and what their strengths and weaknesses are. It also gives you, the writer the opportunity to show your ability to identify, access, understand and critically evaluate relevant material in top rate academic journals.

The literature review must be defined by a guiding concept (e.g., the problem or issue you are discussing or your argumentative research project). **It is not a descriptive list of the material available, or a set of summaries.** It is not recommended to start every paragraph with the name of an author as this may mean that the literature is organised alphabetically or chronologically and not by concepts or topics investigated.

A succession of “cut and paste” article summaries is **NOT** a literature review either!

The following guidelines should be considered when writing the literature review section.

- The literature review should be organised around and related directly to the research project or research question you are developing.
- Any article discussed and analysed in the literature review must have a reference. The full reference should be included in the References section of the project report.
- The literature review should be as current and up to date as is practicable
- Do not cite any reference you have not read.
- The literature review should end with a summary of what is and is not known, should highlight “gaps” in the literature, either acknowledged or discovered, and should formulate questions that need further research

Tips on conducting a Lit Review:

- <http://www.writing.utoronto.ca/advice/specific-types-of-writing/literature-review>
- <http://www.nuigalway.ie/mscsed/download/file/124/>
- https://web.njit.edu/~egan/Writing_A_Literature_Review.pdf

5.4.3 Research Methodology

The aim of the Research Methodology section is to provide details regarding

- the steps followed in the research work (process).
- the data (e.g., database records, tools, software, algorithms, metrics, etc.) you need in order to be able to answer your research question.
- how you get and use those data.
- how the data and the results that are collected or measured in the project are analysed.

Usually, the data requirements are specified in several research objectives. These objectives are a form of data “shopping list” and clarify what amount of data you need to collect.

5.4.4 Design and Solution Development

The report should provide details on technologies, techniques and/or architecture/framework that underlie development and implementation of the solution proposed to answer the research question.

A description of the developed solution should present your technical work. Only the final stage of the developed solution should be presented.

Due to the space limitation (the report should not exceed 20 pages) code should not be included in the report. For example, if an algorithm was proposed, implemented, and tested a pseudocode or word-based steps description of the algorithm may be included. The student can show and discuss the code of the algorithm during the demo run in the viva.

5.4.5 Evaluation and Results Analysis

A comprehensive set of tests should be run by the student to evaluate and validate the proposed solution. However, only the most relevant results should be presented in the report. All figures and tables should be numbered and referred in the text of the report. The meanings of the data (e.g., numbers) presented in the tables and figures should be explained in the report.

The students should also present in the project report a comprehensive analysis of the results the main findings of the study using multiple strategies as well as the implications of these finding both from academic and practitioner perspective are presented.

It is also important to situate findings in the literature. Do your findings support, contradict, or modify the findings of those authors on whom you have drawn your research question?

5.4.6 Conclusion and Discussion

The conclusion and discussion section are a very important parts of the report. It has to briefly reiterate the research question(s) addressed and to summarize the most important outcomes of the research work. Insights gained throughout the development and evaluation of the proposed solution are critically analysed and discussed.

In addition, further avenues for research that could be explored should be discussed.

The relevance of the research output to different stakeholders (e.g., companies, institutions services, end-users, clients, etc.) and potential for commercialization should also be discussed.

5.4.7 References

This should contain a full list of any source **directly referred to or quoted** in the text following the format provided in the research project template.

The NCI library has developed several useful tools to assist students with both their in-text referencing and the presentation of their reference lists. If you are unsure how to reference correctly, please consult these tools.

As the research project report should not have more than 20 pages you must select and refer in the text the most relevant references that related to your research topic and work.

5.4.8 Report Template

The research paper style report should comply with a specific presentation style and format as pre-defined by the report template provided on the Moodle page.

5.5 Configuration Manual

The configuration manual shall describe the required software tools and settings in order to successfully replicate the experimental setup. This manual is submitted separately from the research paper style report and it will not count towards the suggested word count. It **should not** discuss how to install standard software or tools.

5.6 Viva Video Recording

The viva shall involve a presentation of the research work carried out and a demonstration of the results to at least two academic examiners. A demo of the artefact/product developed will be required to be presented in the viva. The learner must submit the videos required for the viva in order to pass the Practicum/ Research Practicum/ Internship module.

Students must submit a viva video recording that provides a presentation on their research work and a demonstration of the results and artefact/product developed. The presentation shall use visual aids as appropriate including slides (PowerPoint, LaTeX Beamer, etc.). A demo of the artefact/product developed is required to be presented in the viva. The learner must submit on module's Moodle page the videos required for the viva in order to pass the Practicum/ Research Practicum/ Internship module.

The video recording(s) should be organised by the student (roughly) as follows:

- 10 min conference-style presentation that encompasses all key aspects of the work, i.e. all sections of the report
- 5-10 min live demo of key artefacts/ components of the evaluation and results

The video recording(s) will be assessed by two examiners, one of the examiners is the supervisor.

The supervisor will provide the student with a set of questions from the examiners in relation to their work. The students must upload their answers on Moodle page before the deadline.

Important Note: The College does reserve the right to request a live viva session to take place in the event where examiners feel that further clarification is required regarding a student submission.

Therefore, ALL students will be required to make themselves available for a possible live viva session that would take place at date specified well in advance on the Moodle page.

Students for which a live viva presentation is required will be notified via a Microsoft Teams meeting invite and should ensure that they accept that invite so that same is added to their Outlook Calendar. A total of 30 minutes will be allocated per student.

6. Plagiarism

Plagiarism arises when extracts from someone else's work are used without acknowledgement, i.e., in the case of written work, not italicised, not set in quotation marks, and not referenced. This is not limited to text but can also include graphics, tables, photographs, video, music, and computer code.

Plagiarism is also submitting the same piece of work for assessment under multiple modules. Plagiarism is not an acceptable practice and will cause a piece of work to be penalised or regarded as null.

The submission of plagiarised materials for assessment purposes is fraudulent and all suspected cases will be investigated and dealt with appropriately by the College following the procedures outlined here and with reference to the Code of Discipline

If perpetrated on a substantial scale, especially with intention, disqualification from an examination or award may occur.

Whilst some cases of plagiarism can arise through poor academic practice with no deliberate intent to cheat, this still constitutes a breach of acceptable practice, and requires to be appropriately investigated, and acted upon.

The official regulations regarding plagiarism are presented in Academic Integrity Policy document available on the NCI Student Portal under [Academic Integrity](https://www.ncirl.ie/Students/Academic-Integrity) (<https://www.ncirl.ie/Students/Academic-Integrity>) and supplementary information is included on School notice boards and appropriate hand-outs to learners.

Students are required to sign a short declaration (See Appendix A – Project Submission Sheet) that the research paper style report submitted by them for assessment purposes is their own. This statement is attached to the report and requires acknowledgment that the student has read and understood the plagiarism regulations.

6.1 Use of AI sources in Teaching and Learning: Student Guide

All work submitted to NCI is subject to NCI's Guidelines on Academic Integrity. According to the National Academic Integrity Network; academic integrity means being true to yourself and being fair to your classmates. The purpose of assessment is to support learning, show student understanding and demonstrate the achievement of specific learning outcomes by that student. It

is the responsibility of the student to ensure all submitted work correctly acknowledges the source of any data which is not the original work of the learner.

Academic staff at NCI use technology such as Turnitin's AI detection, along with heuristic methods such as writing comparisons, language consistency, source checks to detect possible Academic Dishonesty. If a member of staff suspects Academic Dishonesty via AI, you will be subject to the Disciplinary Process of the college.

If you have used AI sources in your work in any capacity, then you must acknowledge, describe, and evidence their use by filling out the AI Referencing Supplement Document (see Appendix A) and attaching it with your submission.

Further instructions on how to refer AI sources can be found on the NCI Library web site.

<https://libguides.ncirl.ie/useofaiinteachingandlearning/studentguide>

7. Library Resources

The Library Help Centre² delivers classes and offers one-to-one consultations on a number of topics including online resources, referencing and conducting literature reviews.

7.1 Information Project Officer

The Information Project Officer offers one-to-one consultations to postgraduate students on a number of topics; if you have a detailed enquiry you are invited to make an appointment. Specific assistance is provided in the areas of (a) accessing the library webpage and using the library catalogue, (b) online databases, (c) research strategies, (d) academic writing, (e) referencing, (f) conducting a literature review and (g) accessing the past exam papers database and theses.

7.2 Citations

All work submitted by students for assessment purposes is accepted on the understanding that it is their own work and written in their own words except where explicitly referenced using the accepted norms and formats of the appropriate academic discipline. The style of referencing to be used is specified in the report template provided on the Moodle page.

Please refer to the How to Reference³ section of the NCI Library for further details.

At the end of the research paper style report, there should be a section marked "References" where all references should be listed in full.

8. Copyright, Intellectual Property and Commercialisation

Copyright in the Practicum/ Research Practicum/ Internship report resides with the student. All other intellectual property (including inventions, discoveries, copyright in computer software (including source and object codes) and other kinds of copyright, semiconductor mask works, trade secrets, know how, research data, tangible research property and similar property/rights)

² <http://libguides.ncirl.ie/staffguide/researchandteachinghelp>

³ <https://www.ncirl.ie/Library/Library-Help-Centre/How-To-Reference>

conceived, invented, or reduced to practice by the student during, or as a result of, research undertaken at the College shall be the property of the National College of Ireland. College policy with respect to ownership and exploitation of intellectual property (IP) is set out in the current NCI Intellectual Property Policy (to which the attention of the candidate is drawn generally). The NCI Intellectual Property Policy document is available on the NCI Support Hub (<https://ncisupporthub.ncirl.ie/hc/en-ie/articles/6592724149276-11-QAES-Research>). Benefits accruing to the College from the subsequent commercialisation of the intellectual property referred shall be allocated in accordance with such policy.

Students that wish to commercialise their research work or a project related technology have the opportunity to explore this path. In this case, students will be required to complete and submit to Programme Director the NCI Invention Discourse Form available from IP Policy Document published on the NCI student portal.

This document aims to identify the potential commercial opportunities, to explore the related markets and to justify the commercialisation of products or services as results of work done for the project.

Technology commercialisation applications will be reviewed by NCI Technology Transfer Office in terms of commercialisation potential. If application approved, NCI will provide protection for company spin-out, protection of the invention and support for negotiation of licence agreement.

9. Deadline Extensions, Exit, Withdrawal and Deferral of the Course

The Practicum/ Research Practicum/ Internship submission date is absolute and final. Applications for extensions will not be tolerated except in line with those circumstances as set forth in the College Regulations. Any student seeking to obtain an extension should first consult the College Regulations.

Occasionally and after consultation with the Programme Director, a student may decide to withdraw or defer their studies prior to completion of their Practicum/ Research Practicum/ Internship. In this instance the student must consult the College Regulations in relation to deferrals and contact the Programme Director to seek guidance on what the consequences of any such action may be.

If the student does not want to complete the Practicum/ Research Practicum/ Internship module may apply to be transfer to the Postgraduate Diploma course. An Exit Award Form (see Appendix B) must be submitted to Registrar. The student must achieve the required credits to exit with the Postgraduate Diploma once the transfer request was approved.

If the student wishes to withdraw from the course or to apply for deferral from the course, he/she must communicate this to the NCI Registrar and the Programme Director.

10. Useful Contacts

Position	Name	E-mail Address
Dean of School of Computing	Prof. Paul Stynes	paul.stynes@ncirl.ie
Vice Dean School of Computing	Prof Cristina Muntean	cristina,muntean@ncirl.ie
Programme Director MSc Artificial Intelligence	Dr Abdul Shahid (January intake) Dr Anh Duong Trinh (September intake)	Abdul.Shahid@ncirl.ie anhduong.trinh@ncirl.ie
Programme Director MSc AI for Business	Dr Anu Sahni (September and January intakes)	anu.sahni@ncirl.ie
Programme Director MSc TopUp Artificial Intelligence	Dr Anh Duong Trinh	anhduong.trinh@ncirl.ie
Programme Director MSc Cyber Security (full-time)	Dr Arghir Nicolae Moldovan (Group A) Dr Rohit Verma (Group B)	arghir.moldovan@ncirl.ie rohit.verma@ncirl.ie
Programme Director MSc Cybersecurity (part-time)	Dr Rohit Verma	rohit.verma@ncirl.ie
Programme Director MSc TopUp Cybersecurity	Dr Abdul Shahid	Abdul.Shahid@ncirl.ie
Programme Director MSc Fintech	Dr Rohit Verma	Rohit.Verma@ncirl.ie
Programme Director MSc in Data Analytics – SEPT 24 & onward intakes (Full-time)	Mr Michael Bradford (Group A) Dr Anu Sahni (Group B) Dr Giovani Estrada (Group C)	michael.bradford@ncirl.ie anu.sahni@ncirl.ie giovani.estrada@ncirl.ie
Academic Operations ticket system		academicoperations@ncirl.ie
Project Information Officer	Mr Keith Brittle	Keith.brittle@ncirl.ie
Registrar's Office		registrarsoffice@ncirl.ie
Library		library@ncirl.ie
Student Services		studentservices@ncirl.ie
Exams Office		examinations@ncirl.ie
International Office	Ms Sheila Mahon	sheila.mahon@ncirl.ie
SoC Ethics Filter Committee	Prof Cristina Muntean	cristina.muntean@ncirl.ie

11. Appendix A - Project Submission Sheet

National College of Ireland

School of Computing



Student Name:

Student ID:

Programme: **Year:**

Module:

Lecturer:

Submission Due Date:

Project Title:

Word Count: **Page Count:**.....

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the bibliography section. Students are required to use the Referencing Standard specified in the report template. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

I agree to an electronic copy of my thesis being made publicly available on NORMA the National College of Ireland's Institutional Repository for consultation.

Signature:

Date:

PLEASE READ THE FOLLOWING INSTRUCTIONS AND CHECKLIST

Attach a completed copy of this sheet to each project (including multiple copies)	☐
Attach a Moodle submission receipt of the online project submission, to each project (including multiple copies).	☐
You must ensure that you retain a HARD COPY of ALL projects, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer.	☐

Office Use Only	
Signature:	
Date:	
Penalty Applied (if applicable):	

AI Acknowledgement Supplement

[Insert Module Name]

[Insert Title of your assignment]

Your Name/Student Number	Course	Date

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]	
[Insert Description of use]	
[Insert Sample prompt]	[Insert Sample response]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

12. Appendix B - Exit Award Form

AWARD EXIT / TRANSFER FORM

This form should be used if you wish to transfer from one programme to another or if you wish to exit with an award from the National College of Ireland. The completed form must be submitted to the Registrar's Office, National College of Ireland, Mayor Street, IFSC, Dublin 1.

Name:

Address:

.....

.....

Student Number:

I wish to transfer from my current programme

Transfer to [Programme Title]:

Transfer to [Programme Code]:

Year:

OR

I wish to apply for an exit award from:

Exit Award [Programme Title]:

Exit Award [Programme Code]:

Year:

Summarise briefly what programme you are applying to exit from / transfer to:

.....

.....

Student Signature: **Date:**

ECTS Credits have been accumulated [Tick Box]: **Yes** **No**

Internal Approval: **Date:**

13. Appendix C – Declaration of Ethics Consideration Form

National College of Ireland



DECLARATION OF ETHICS CONSIDERATION

School of Computing

Student Name:

Student ID:

Programme **Year:**

Module:

Project Title:

Please circle (or highlight) as appropriate

This project involves human participants	Yes / No
--	----------

Introduction

Secondary data refers to data that is collected by someone other than the current researcher. Common sources of secondary data for social science include censuses, information collected by government departments, organizational records and data originally collected for other research purposes. **Primary data**, by contrast, is collected by the investigator conducting the research.

Target system refers to any asset, device, infrastructure component, hosting environment, network, or application, that is intended to be used by the current researcher as the target system for testing their ICT solution artefact. This can include a penetration testing target or any system under test.

Rules of Engagement is a document that contains the formal approvals, authorizations, scope, and other general guidelines or formal objectives necessary to execute a penetration testing engagement.

The following decision table will assist you in deciding if you have to complete the Declaration of Ethics Consideration Form or/and the Ethics Application Form.

Public Data/Target System	Y	Y	Y	Y	N	N	N	N
Private Data/Target System	Y	Y	N	N	Y	Y	N	N
Human Participants	Y	N	Y	N	Y	N	Y	N
Declaration of Ethics Consideration Form	X	X	X	X	X	X	X	
Ethics Application Form	X		X		X		X	

A project that does not involve human participants requires ONLY completion of Declaration of Ethics Consideration Form and submission of the form on the module's Moodle page.

A project that involves human participants requires ethical clearance. Thus, it requires the completion of both a Declaration of Ethics Consideration Form AND an Ethics Application Form. These must be submitted through the module's Moodle page. Please refer to and ensure compliance with the ethical principles stated in NCI Ethics Form available on the Moodle page.

Please circle (or highlight) as appropriate.

The project makes use of secondary dataset(s) or target system(s) created by the researcher	Yes / No
The project makes use of public secondary dataset(s) or target system(s)	Yes / No
The project makes use of non-public secondary dataset(s) or target system(s) Approval letter from non-public secondary dataset(s) or target system(s) owner received	Yes / No Yes / No

Sources of Data/ Target System

It is students' responsibility to ensure that they have the correct permissions/authorizations to use any data or target system in a study. Projects that make use of data or target system that does not have authorization to be used, will not be graded for that portion of the study that makes use of such data or target system. For target systems, the student must ensure that the asset is used in accordance with the company's acceptable usage policies and Terms of Use.

Public Data / Public Target System

A project that makes use of public secondary dataset(s) or public target system(s) **does not need ethics permission** but **needs a letter/email from the copyright holder/ target system owner** regarding potential use.

Some websites and data sources allow their data sets or platforms to be used under certain conditions. In these cases, a letter/email from the copyright holder is NOT necessary, but the researcher should cite the source of this permission and indicate under what conditions the data or target system are allowed to be used. See [Appendix I](#) for examples of permissions for data granted by Fingal Open Data, and Eurostat website. See [Appendix III](#) for examples of permissions granted by some penetration testing target systems.

Where websites or data sources indicate that they do not grant permission for data or their platform to be used, you will still need a letter/email from the copyright holder. For example, see [Appendix II](#) for an example from the Journal of Statistics Education.

Private Data / Private Target System (not created by the researcher)

A project that makes use of non-public (private) secondary dataset(s) or target system must receive usage permission from the School of Computing.

An approval letter/email from the owner (e.g. institution, company, etc.) of the non-public secondary dataset or target system must be attached to the Declaration of Ethics Consideration. For datasets, the letter/email must confirm that the dataset is anonymised and permission for data processing, analysis and public dissemination is granted.

For target systems, the letter/email must provide evidence that permission/authorization has been granted to the student for using the system. In addition to this, in the cases where the goal of the project is to conduct penetration testing, vulnerability scans, or other ethical hacking engagement, the student must also provide a document describing the rules of engagement.

Evidence to include for use of secondary dataset(s) / target system(s)

Include dataset(s) / target system(s) owner letter/email or cite the source for usage permission.

For dataset(s)

1. Name of the dataset, a brief description of its nature
2. citation of the paper if required by the authors
3. URL of the location of the dataset
4. License of usage

For target environment(s)

1. Name of the target environment (platform/application/websites), a brief description of

CHECKLIST

Non-public/private secondary dataset(s)/target system(s)	
- Dataset / target system Owner letter/email	Yes / No
- Rules of Engagement if applicable see <u>Appendix IV</u> - Rules of Engagement Worksheet) is attached to this form	Yes / No
OR	
Citation and link to the website where permission is granted – provided in this form	Yes / No

ETHICS CLEARANCE GUIDELINES WHEN HUMAN PARTICIPANTS ARE INVOLVED

The Ethics Application Form must be submitted on Moodle for approval prior to The Ethics Application Form must be submitted on Moodle for approval prior to conducting the work.

Considerations in data collection

- Participants will not be identified, directly or through identifiers linked to the subjects in any reports produced by the study
- Responses will not place the participants at risk of professional liability or be damaging to the participants' financial standing, employability or reputation
- No confidential data will be used for personal advantage or that of a third party

Informed consent

- Consent to participate in the study has been given freely by the participants
- participants have the capacity to understand the project goals.
- Participants have been given information sheets that are understandable
- Likely benefits of the project itself have been explained to potential participants
- Risks and benefits of the project have been explained to potential participants
- Participants have been assured they will not suffer physical stress or discomfort or psychological or mental stress
- The participant has been assured s/he may withdraw at any time from the study without loss of benefit or penalty
- Special care has been taken where participants are unable to consent for themselves (e.g children under the age of 18, elders with age 85+, people with intellectual or learning disability, individuals or groups receiving help through the voluntary sector, those in a subordinate position to the researcher, groups who do not understand the consent and research process)
- Participants have been informed of potential conflict of interest issues
- The onus is on the researcher to inform participants if deception methods have to be used in a line of research

I have read, understood, and will adhere to the ethical principles described above in the conduct of the project work.

Signature : _____

Date : _____

Examples of source citations

APPENDIX I

1) Fingal Open Data: <http://data.fingal.ie/About>

Licence

Citizens are free to access and use this data as they wish, free of charge, in accordance with the Creative Commons Attribution 4.0 International License (CC-BY).

Note: From November 2010 to July 2015, data on Fingal Open Data was published in accordance with the PSI general licence.

Use of any published data is subject to Data Protection legislation.

Licence Statement

Under the CC-BY Licence, users must acknowledge the source of the Information in their product or application by including or linking to this attribution statement: "Contains Fingal County Council Data licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence".

Multiple Attributions

If using data from several Information Providers and listing multiple attributions is not practical in a product or application, users may include a URI or hyperlink to a resource that contains the required attribution statements.

2) Eurostat:

<https://ec.europa.eu/eurostat/about/policies/copyright>

COPYRIGHT NOTICE AND FREE RE-USE OF DATA

Eurostat has a policy of encouraging free re-use of its data, both for non-commercial and commercial purposes. All statistical data, metadata, content of web pages or other dissemination tools, official publications and other documents published on its website, with the exceptions listed below, can be reused without any payment or written licence provided that:

- the source is indicated as Eurostat
- when re-use involves modifications to the data or text, this must be stated clearly to the end user of the information

APPENDIX II

3) Journal of Statistics Education: http://jse.amstat.org/jse_users.htm

JSE Copyright and Usage Policy

Unlike other American Statistical Association journals, the Journal of Statistics Education (JSE) does not require authors to transfer copyright for the published material to JSE. Authors maintain copyright of published material. Because copyright is not transferred from the author, permission to use materials published by JSE remains with the author. Therefore, to use published material from a JSE article the requesting person must get approval from the author.

APPENDIX III

1) Test Site for Acunetix Web Vulnerability Scanner. <http://testphp.vulnweb.com/>

Warning: This is not a real shop. This is an example PHP application, which is intentionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip: Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more.

2) The AltoroJ website. <http://testfire.net/>

The AltoroJ website is published by IBM Corporation for the sole purpose of demonstrating the effectiveness of IBM products in detecting web application vulnerabilities and website defects. This site is not a real banking site. Similarities, if any, to third party products and/or websites are purely coincidental. This site is provided "as is" without warranty of any kind, either express or implied. IBM does not assume any risk in relation to your use of this website. For more information, please go to <http://www-142.ibm.com/software/products/us/en/subcategory/SWI10>.

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1) TryHackMe. <https://tryhackme.com/r/legal/terms-of-use> and <https://tryhackme.com/r/legal/acceptable-use-policy>

2. Using the Site

2.1 The Site is for your personal use only. You must not use it for business purposes or provide access to it through your account to any other person.

2.2 As a condition of using the Site, you must not, except as permitted by the rules of specific Rooms and only to the extent of within a Room:

a. misuse or attack the Site or any Room by introducing viruses, trojans, worms, logic bombs or any other material which is malicious or technologically harmful to the Site, other users, or the communications networks and infrastructure used to access and make available the Site (such as by way of a denial-of-service attack);

b. attempt to gain unauthorized access to the Site and/or Room, the server on which the Site is stored or any server, computer or database connected to the Site; and

c. engage in any unlawful, inappropriate and abusive use of the content available on the Site at all times.

2.3 We may prevent or suspend your access to the Site if you fail comply with the Agreement or any applicable law.

2.4 Users without subscriptions can only access non-subscription Rooms.

APPENDIX IV

Rules of Engagement: <https://www.sans.org/posters/pen-test-rules-of-engagement-worksheet/>

Rules of Engagement Worksheet:

Penetration Testing Team Contact Information:

Primary Contact: _____

Email: _____

Mobile Phone: _____

Secondary Contact: _____

Email: _____

Mobile Phone: _____

Target Organization Contact Information:

Primary Contact: _____

Mobile Phone: _____

Email: _____

Secondary Contact: _____

Mobile Phone: _____

Email: _____

"Daily Debriefing" Frequency: _____

"Daily Debriefing" Time/Location: _____

Start Date of Penetration Test: _____

End Date of Penetration Test: _____

Testing Occurs at Following Times: _____

Will test be announced to target personnel: _____

Will target organization shun IP addresses of attack systems: _____

Does target organization's network have automatic shunning capabilities that might disrupt access in unforeseen ways (i.e. create a denial-of-service condition), and if so, what steps will be taken to mitigate the risk:

Would the shunning of attack systems conclude the test: _____

If not, what steps will be taken to continue if systems get shunned and what approval (if any) will be required:

IP addresses of penetration testing team's attack systems:

Is this a "black box" test: _____

What is the policy regarding viewing data (including potentially sensitive/confidential data) on compromised hosts:

Will target personnel observe the testing team: _____

Signature of Primary Contact representing Target Organization

Date _____

Signature of Head of Penetration Testing Team

Date _____

If necessary, signatures of individual testers:

Signature _____
Date _____

Signature _____
Date _____

Signature _____
Date _____

14. Appendix D - Human Participants Ethical Review Application Form

National College of Ireland Ethical Guidelines and Procedures for Research involving Human Participants



JANUARY 2023

1. Introduction

All research involving human participants that is conducted by students or staff at the National College of Ireland should be done so in an ethical manner. The college has therefore developed an Ethics Committee, which acts as a sub-committee of the Research Committee, to ensure that ethical principles pertaining to research involving human participants are upheld and adhered to. All researchers intending to use human participants as part of their projects are thus required to reflect upon any potential ethical issues and submit their research proposals for ethical review before commencing data collection.

This document gives an overview of the core ethical principles guiding research in NCI, while also documenting the procedures required for seeking ethical approval of research involving human participants.

Am I conducting research?

Research is defined as “the attempt to derive generalisable new knowledge by addressing clearly-defined questions with systematic and rigorous methods” (NHS Health Research Authority). Sometimes, we collect data in order to research a service or practice we are engaged in (“service evaluation”). The main difference between research and research of a service - is in the aim: research is trying to create new generalisable knowledge, and research related to service evaluation is trying to evaluate whether a delivered service/practice is working well. One project may have both aims included in it. It can be confusing if a service or intervention is involved, whether or not research is being conducted. If new or competing interventions are being evaluated, then it is likely to be research, whereas if an existing service is being conducted anyway, with an evaluative component, then it is likely to be a service evaluation. Research requires consideration of the below guiding principles

Guiding Principles

In line with other research institutions, there are three core guiding principles governing the ethical conductance of research involving human participants at NCI. These principles stem from the *Belmont Report* (1979) published by the National Commission for the Protection of Human Subjects of Biomedical and Behavioural Research. While it is recognised that these principles may be operationalised differently depending on the specific research discipline, it is recommended that these are consulted as a starting point for any research involving human participants.

1.1 Principle 1: Respect for Persons

This principle entails recognition that participants should be treated as autonomous individuals and hence should never be coerced or swayed into participating in a research project against their will. The participant’s right to withdraw from a research study at any time should be respected, as well as their right to dignity and protection from harm.

Respect for individuals can often be implemented in practice via the process of informed consent, whereby potential participants are made fully aware of the requirements involved in participation. While it is recognised that in certain cases deception (i.e. the withholding of certain information from participants) may take place, this should only occur when it is robustly justified for the validity of the research. In cases where deception is justified, researchers should ensure that any potential risk resulting from this measure is minimised. Participants should also be fully debriefed on the nature of the research after it has taken place.

The principle of respect also requires researchers to protect individuals from vulnerable groups who may have diminished autonomy (see section 4.2 for more detail as to what constitutes vulnerable groups). Where full informed consent is not possible for such population groups, consent may instead be sought from their guardians. In all cases however clear assent, or willingness to participate, should be demonstrated from participants.

1.2 Principle 2: Beneficence and non-maleficence

This principle specifically focuses on the need to protect the well-being of participants. Any potential risk to participants should be minimised, whether that be risk of physical discomfort or of any psychological, emotional or social distress, while possible benefits should be maximised. Researchers adhering to this principle should thus ensure that any potential benefits derived from carrying out the study (e.g. in terms of knowledge gained) should outweigh potential risks. Even in cases where there is only a slight potential risk of harm, participants should be provided with appropriate support to alleviate this.

1.3 Principle 3: Justice

This principle emphasises the need to employ fairness in the distribution of benefits and risks to participants. The way in which participants are selected to take part in research should relate to the purpose of the study, as opposed to other factors such as availability or manipulability of participants. The exploitation of vulnerable populations should be avoided.

Where applicable, researchers are encouraged to consult guidelines stemming from their own professional bodies (e.g. The Psychological Society of Ireland) in addition to the general guiding principles above when planning their research. Researchers should also be sensitive to those issues which are specific to the population under investigation and the methodology that is employed in the project (e.g. qualitative methodologies involving the recording of data may raise issues relating to participants' right to anonymity, as well as the ethical management and use of data). Detailed consideration should be given to all these issues when planning research and when completing the Ethical Review Application form.

2. Ethics Committee

The NCI Ethics Committee was established by the Academic Council in 2012. Acting as a sub-committee to the Research Committee, its role is to oversee ethical issues arising from all research involving human participants that is conducted by students and staff of the college. The key purpose of this committee is to safeguard against any potential harm to participants, and to ensure that their rights are recognised in line with the guiding principles outlined above.

The Ethics Committee reviews all research proposals posing ethical risk to the participants involved, however the decision as to whether projects pose ethical risk is firstly made via the appropriate Filter Committee which operates at School level (see organisational structure in Figure 1 below). The Filter Committees may review and approve research proposals which are of low ethical risk, while referring those of high ethical risk to be considered by the Ethics Committee (see categories of ethical risk in section 4.1).

While the Filter Committees are made up of staff members with subject-specific knowledge, membership of the Ethics Committee should comprise of no less than five representatives from both the School of Computing and the School of Business, including representatives from the Research Committee.



Figure 1: Committee Structures.

3. Review Process

Any staff or student of NCI wishing to conduct a study involving human participants should first submit the Ethical Review Application Form or, in certain limited situations, the Ethical Review Exemption Form (both included at the end of this document), to the relevant School Filter Committee at proposal stage. This initial review will result in a graded categorisation of ethical risk, as outlined below.

3.1 Categorisation of Ethical Risk

Research category A

Research in this category poses little ethical risk to the participants involved. Specifically, it refers to research involving human volunteers, but **excluding** studies involving:

- therapeutic interventions
- new research methodologies
- vulnerable populations (see section 4.2)
- deception of the participants
- any other significant physical, social or psychological risk to participants

Research category B

Research in this category involves human volunteers **including** studies involving:

- therapeutic interventions
- new research methodologies
- vulnerable populations (see section 4.2)
- deception of the participants

- any potentially significant risk to participants

Research Category C

This specifically refers to research involving human volunteers who are service users, patients, staff, records, etc., within the sphere of the HSE or similar setting (but not including clinical trials of investigative medicinal products).

3.2 Vulnerable groups

There are a number of participant populations that may fall under the heading of ‘vulnerable groups’. These groups require consideration of unique ethical challenges regardless of the nature of the project. Research involving such populations should therefore always be reviewed by the Ethics Committee.

Groups that may be classed as vulnerable include, but are not limited to:

- Children (under 18 years of age)
- The very elderly
- People with an intellectual or learning disability
- Individuals or groups receiving help through the voluntary sector
- Those in a subordinate position to the researcher (e.g. employees)
- Any other groups who might not understand the research and consent process

Note: in addition to the Ethical Review process, any researchers intending to work directly with children will be required to undergo Garda Vetting in advance of the proposed research.

3.3 Exemption from Full Ethical Review

In certain limited cases, researchers can apply for an exemption from full ethical review. In such cases, the Ethical Review Exemption form should be completed, explicitly detailing why the exemption is sought.

In completing this form, researchers must declare that the research does not involve any of the following:

- Vulnerable groups
- Sensitive topics
- Risk of psychological or mental distress
- Risk of physical stress or discomfort
- Any other risk to participants
- Use of drugs or invasive procedures (e.g. blood sampling)
- Deception or withholding of information from participants
- Conflict of interest issues
- Access to data by individuals or organisations other than the researchers
- Any other ethical dilemmas

3.4 Outcomes of Review Process

Following consideration of research projects submitted for Ethical Review, each Filter Committee will submit a report to the Ethics Committee summarising the applications considered and the decisions made.

For research that is deemed to fall under Research Category A (low ethical risk), a favourable outcome at the relevant Filter Committee will be sufficient to secure ethical approval. Research falling under the other two categories must however be considered by the Ethics Committee before approval may be granted.

On the basis of this review, four key outcomes may arise:

1. Research proposal approved (no recommendations)
2. Research proposal approved pending minor revisions (to be accepted by the Chair and Research Supervisor)
3. Research proposal approved pending major revisions (to be resubmitted and approved by the Ethics Committee)
4. Research proposal rejected (resubmission necessary)

A summary of the processes involved in applying for ethical approval can be seen in Figure 2.

Appeals

Appeals against the Committee's decision may be made within ten working days. In this case, at least three members of the Ethics Committee, none of whom will have reviewed the initial application, may review this along with any additional information submitted by the applicant.

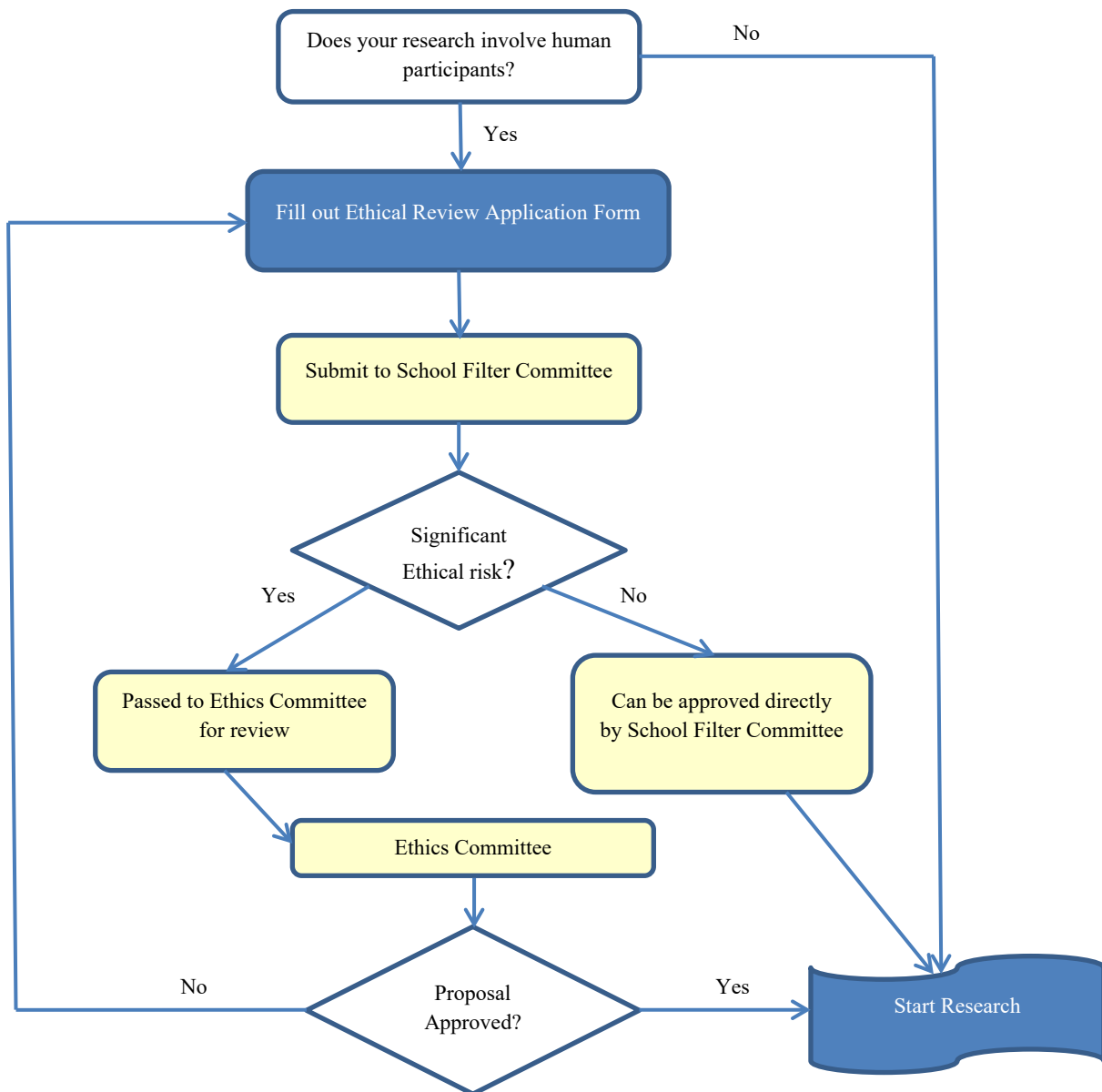


Figure 2: Process chart for seeking Ethical Approval

Ethics Application Checklist

To be submitted alongside three printed and signed copies of the ethics application.

Please complete the below checklist, ticking each item to confirm that it has been addressed.

1. I agree to obtain informed written consent from all human participants aged over 18 who are involved in this research (or if circulating digitally, I will ensure that informed consent is completed, and will have the participants indicate their informed consent by continuing with their study engagement).	☐
2. I agree to obtain informed written consent from the parents of anyone aged under 18 in this research (or from the schools if appropriate), and informed written assent from those under 18 in this research.	☐
3. I append a letter of agreement from a clinically responsible individual agreeing to (where appropriate) help me recruit/provide clinical support in the event that participants become distressed/host the study data collection.	☐
4. I append a letter of agreement from an external institution or organisation agreeing to host the study.	☐
5. I agree to comply with NCI's Data Retention Policy.	☐
6. I have appended a) information sheet, b) consent form/assent form, c) debriefing sheet.	☐
7. I have provided details of how non-anonymised data will be stored, in a safe and encrypted manner.	☐
8. I have included my contact details and those of my supervisor (where appropriate). I have only included my NCI email address and not included any personal contact information.	☐
9. I have given sufficient details on the proposed study design, methodology, and data collection procedures, to allow a full ethical review, and I understand that my failure to give sufficient detail may result in a resubmission being required.	☐
10. I understand that if I make changes to my study following ethical approval, it is my responsibility to seek an ethics amendment if the change merits ethical consideration.	☐

National College of Ireland

Human Participants Ethical Review Application Form

All parts of the below form must be completed. However in certain cases where sections are not relevant to the proposed study, clearly mark NA in the box provided.

Part A: Title of Project and Contact Information

Name

Student Number (if applicable)

Email

Status:

Undergraduate ☐

Postgraduate ☐

Staff ☐

Supervisor (if applicable)

Title of Research Project

Category into which the proposed research falls (see guidelines)

Research Category A ☐

Research Category B ☐

Research Category C ☐

Have you read the NCI Ethical Guidelines for Research with Human Participants?

Yes ☐

No ☐

Please indicate any other ethical guidelines or codes of conduct you have consulted

Has this research been submitted to any other research ethics committee?

Yes ☐

No ☐

If yes please provide details, and the outcomes of this process, if applicable:

Is this research supported by any form of research funding?

Yes ☐
No ☐

If yes please provide details, and indicate whether any restrictions exist on the freedom of the researcher to publish the results:

Part B: Research Proposal

Briefly outline the following information (not more than 200 words in any section).

Proposed starting date and duration of project

The rationale for the project

The research aims and objectives

The research design

The research sample and sample size

Please indicate the sample size and your justification of this sample size. Describe the age range of participants, and whether they belong to medical groups (those currently receiving medical treatment, those not in remission from previous medical treatment, those recruited because of a previous medical condition, healthy controls recruited for a medical study) or clinical groups (those undergoing non-medical treatment such as counselling, psychoanalysis, in treatment centres, rehabilitation centres, or similar, or those with a DSM disorder diagnosis).

If the study involves a **MEDICAL** or **CLINICAL** group, the following details are required:

- a) Do you have approval from a hospital/medical/specialist ethics committee?
If YES, please append the letter of approval. Also required is a letter from a clinically responsible authority at the host institution, supporting the study, detailing the

support mechanisms in place for individuals who may become distressed as a result of participating in the study, and the potential risk to participants.

If NO, please detail why this approval cannot or has not been sought.

b) Does the study impact on participant's medical condition, wellbeing, or health?

If YES, please append a letter of approval from a specialist ethics committee.

If NO, please give a detailed explanation about why you do not expect there to be an impact on medical condition, wellbeing, or health.

The nature of any proposed pilot study. Pilot studies are usually required if a) a new intervention is being used, b) a new questionnaire, scale or item is being used, or c) established interventions or questionnaires, scales or items are being used on a new population. If no such study is planned, explain why it is not necessary.

--

The methods of data analysis. Give details here of the analytic process (e.g. the statistical procedures planned if quantitative, and the approach taken if qualitative. It is not sufficient to name the software to be used).

--

Study Procedure

Please give as detailed an account as possible of a participant's likely experience in engaging with the study, from point of first learning about the study, to study completion. State how long project participation is likely to take, and whether participants will be offered breaks. Please attach all questionnaires, interview schedules, scales, surveys, and demographic questions, etc. in the Appendix.

--

Please identify any ethical issues or risks of harm or distress which may arise during the proposed research, and how you will address this risk. Here you need to consider the potential for physical risk, social risk (i.e. loss of social status, privacy, or reputation), outside of that expected in everyday life, and whether the participant is likely to feel distress as a result of taking part in the study. Debriefing sheets must be included in the appendix if required. These should detail the participant's right to withdraw from the study, the statutory limits upon confidentiality, and the obligations of the researcher in relation to Freedom of Information legislation. Debriefing sheets should also include details of helplines and avenues for receiving support in the event that participants become distressed as a result of their involvement in this study.

Do the participants belong to any of the following vulnerable groups?
(Please tick all those involved).

- ☐ Children;
- ☐ The very elderly;
- ☐ People with an intellectual or learning disability
- ☐ Individuals or groups receiving help through the voluntary sector
- ☐ Those in a subordinate position to the researchers such as employees
- ☐ Other groups who might not understand the research and consent process
- ☐ Other vulnerable groups

How will the research participants in this study be selected, approached and recruited? From where will participants be recruited? If recruiting via an institution or organisation other than NCI please attach a letter of agreement from the host institution agreeing to host the study and circulate recruitment advertisements/email etc.

What inclusion or exclusion criteria will be used?

How will participants be informed of the nature of the study and participation?

Does the study involve deception or the withholding of information? If so, provide justification for this decision.

What procedures will be used to document the participants' consent to participate?

Can study participants withdraw at any time without penalty? If so, how will this be communicated to participants?

If vulnerable groups are participating, what special arrangements will be made to deal with issues of informed consent/assent?

Please include copies of any information letters, debriefing sheets, and consent forms with the application.

Part D: Confidentiality and Data Protection

Please indicate the form in which the data will be collected.

☐ Identified ☐ Potentially Identifiable ☐ De-Identified

What arrangements are in place to ensure that the identity of participants is protected?

Will any information about illegal behaviours be collected as part of the research process? If so, detail your consideration of how this information will be treated.

Please indicate any recording devices being used to collect data (e.g. audio/video).

Please describe the procedures for securing specific permission for the use of these recording devices in advance.

Please indicate the form in which the data will be stored (Please read the NCI Research Data Retention Policy)

☐ Identified ☐ Potentially Identifiable ☐ De-Identified

Who will have responsibility for the data generated by the research?

Is there a possibility that the data will be archived for secondary data analysis? If so, has this been included in the informed consent process? Also include information on how and where the data will be stored for secondary analytic purposes.

If not to be stored for secondary data analysis, will the data be stored for 5 years and then destroyed, in accordance with NCI policy?

☐ Yes

☐ No

Dissemination and Reporting

Please describe how the participants will be informed of dissemination and reporting (e.g. submission for examination, reporting, publications, presentations)?

If any dissemination entails the use of audio, video and/or photographic records (including direct quotes), please describe how participants will be informed of this in advance.

Part E: Signed Declaration

I confirm that I have read the NCI Ethical Guidelines for Research with Human Participants, and agree to abide by them in conducting this research. I also confirm that the information provided on this form is correct.

Signature of Applicant ____

Date _____

Signature of Supervisor (where appropriate):

Date _____

Any other information the committee should be aware of?

15. Appendix E – Internship Contract Tri-party

Internship Contract – Tri party

NATIONAL COLLEGE OF IRELAND

Contract for Internship

1 General

This contract regulates the modalities of the internship which forms part of the Masters programme at National College of Ireland.

2 Responsibilities of the contracting parties

1) The Company commits itself to:

- a) Host the student _____ (student name) for a three-month period between _____ (day/month/year) and _____ (day/month/year)
- b) Provide induction training including an outline of Health and Safety and emergency arrangements.
- c) Draw up the goals and job description for the internship in collaboration with The National College of Ireland.
- d) Provide adequate supervision and feedback for the Student during the internship.
- e) Sign the students monthly learning report.
- f) Allow a member of NCI academic staff to attend a company site visit to discuss student progress during the internship.

2) NCI commits itself to:

- a) Provide the Student with the necessary information on the internship.
- b) Provide adequate support for the student during the internship.
- c) Make the Student aware of the internship code of practice and their rights and responsibilities during the internship.

3) The Student commits herself/himself

- a) To make the most of the learning opportunities which the internship offers.
- b) To carry out carefully the tasks which he/she is assigned according to the goals and contents of the internship drawn up in advance.
- c) To follow the instructions of his/her Mentor in the Company.
- d) To follow the official rules and regulations of the Company and to inform the Company immediately in case of absence (such as in the case of sickness).
- e) To adhere to the Company's pledge of confidentiality and to commercial sensitivity (e.g., copyright) to the same extent as any employee of the firm.

3. Supervision

The Company will assign _____ (name) _____ (title), telephone _____ email _____ as the internship supervisor.

4. Remuneration (if applicable)

The Student will receive a salary of € _____ per week.

5. Working hours

The Student will be required to work from _____(am) to _____(pm)

6. Title of Role Student will have on internship: _____

7. Termination of the Contract

The contract can be terminated for reasons including breach of discipline, unsatisfactory internship or internship conditions, inability to carry out the internship without prior notice by both the Company and the Student after consultation with the National College of Ireland.

8. Contract Copy

Three copies of this contract must be signed. Each contracting party will receive one copy.

Signature on behalf of Company

Printed name (Capitals)

Signature on behalf of NCI

Printed name (Capitals)

Signature of student

Printed name (Capitals)

16. Appendix F – Learner contract/ Code of good conduct

MSc Internship - Learning Contract

Prior to Placement we require that you:

- Read the Internship handbook and sign and submit online on Moodle the Learner Contract/Code of Good Conduct (see Appendix E)
- Meet all deadlines as indicated by the Internship Advisor.
- Provide a CV within the deadline as set by the Internship Advisor.
- To attend all Placement classes/appointments arranged or to notify the Internship Advisor beforehand of unavailability.
- To be available for work from May/June to August or September to December
- Respond to Internship advertisements within 24 hours.
- Acknowledge that Internship is a joint effort – you will need to make sufficient number and variety of applications to ensure you achieve a placement by the required date.
- Ensure you properly researched the vacancies and companies before interview.
- To keep the Internship Advisor informed of any other Internship arrangements made directly by you and ensure the placement has been validated by the Internship Advisor and Programme Director.
- To attend all interviews arranged by and agreed with, the Internship Advisor.
- To accept the first firm offer of employment received on your behalf by the Internship Advisor.
- If you have any concerns about your proposed host employer, you should raise these in advance with the Internship Advisor.
- Before arriving at the host employer, you must undertake any programme of preparation offered by the College and attend pre-departure preparation or briefing meetings of which you are notified.

In order to market Internship to prospective employers, the Internship advisors and National College of Ireland give an assurance in good faith that our students will act in a professional and conscientious manner at all times during their Internship. The expectation is that attendance will be regular and punctual and that grooming, and dress will be appropriate. Every year, NCI students participating in Internship have demonstrated these qualities and attributes and indeed, have in many cases exceeded the expectations of employers. We hope that this tradition will continue long into the future. It is important to remember therefore, that throughout the process you are not only representing yourself, but also National College of Ireland and you are paving the way for future generations of Internship students.

ON Placement, You are required:

- To contact the Internship Advisor at NCI to confirm that you have commenced your Internship). Contact by email within 2 days of your start date with your placement contact details. Give your work phone and email as well as your work supervisor's phone and email details.
- Students must act within all company policies and procedures of their employer.
- Please be aware of the company policy governing the use of email and the Internet. Your placement will be terminated for breaches of company policy in this regard.
- Please ensure you are aware of the company HR policies and have reviewed their handbook.
- Please be always professional and respectful in your communications with your Internship employer. Text speak, emails with slang terms or abbreviations are not acceptable in a work environment.
- Swear words are not acceptable within a work environment.
- Perfect time-keeping and attendance is expected by all employers and by NCI.
- Students are required to be well groomed and dressed appropriately for work.
- Every student is **A REPRESENTATIVE of NCI** within their employer's workplace. It cannot be stressed strongly enough to always behave in an appropriate and professional manner. One must also be aware that the availability of future placement positions by employers will depend largely on the impression left by you.

- Keep the Internship Advisor and academic supervisor informed immediately of any problems that cannot be resolved at a local level or issues relating to the quality of the placement.
- It is best to discuss firstly with your work supervisor and then if needed with your Internship Advisor and academic supervisor. They will talk things through with you and advise you on how best to proceed. The aim is to assist you in any way we can. **It is not acceptable to leave an employer without prior permission from your academic supervisor. If the student leaves the employer without NCI approval, he/she will fail the Internship module.**
- Students who have to sit repeat exams are required to take time out of their company holidays. Please discuss any required time off with your immediate company supervisor.
- Complete and return end of placement feedback questionnaires.
- The College does not accept liability for loss and/or damage to personal property, and students are required to make their own insurance arrangements in this respect.
- The College does not accept liability for third party claims arising out of the use by students of their own vehicles for course trips and travel to placements. Students must therefore inform their own individual insurance company/broker and have policies adjusted to reflect vehicle use outside the standard "social and domestic" cover.
- To monitor your progress the College will maintain contact with you (through email or other channels) during your Internship.
- Students will be required to submit their weekly reports on Moodle
- You may also be required to attend an interview to provide feedback on your Internship or to share your experiences whilst on work experience with the next year's students.

What you can expect of us:

- An initial briefing on the placement process, inclusive of Guidelines, key dates and assessment.
- Presentation and advice on the production of CV's
- Follow up CV's clinics, to examine the structure, language and presentation of your CV and to highlight mistakes in grammar, punctuation and spelling. WE WILL NOT WRITE OR REWRITE CV's.
- Presentation and advice on Interview Techniques.
- Mock Interviews where appropriate to support students with interview process.
- Appropriate support and guidance in students finding their own placements.
- Pre-placement talk on preparing students for placement.
- On-going support while on placement.
- The Internship advisor will act as point of contact with the student while on placement.
- Your academic supervisor will be assigned at the beginning of the Internship, and you will have to attend weekly supervision classes as per the timetable.

I have read, understand, and agree to abide by the above conditions.

Signed:.....

Name: (*printed*).....

Date:.....

17. Appendix G – Secure Own Internship Form

Secure Own MSc Internship Form



If you secure your own internship you must e-mail this form AND a job description to your Career Advisor AND Programme Director for approval.

Student Name: _____
Student Number: _____

Employer Details:

Company name: _____

Company address: _____

Supervisor name: _____

Supervisor title: _____
(must be I.T.)

Contact email: _____

Contact number: _____

Job Details:

Title of position: _____

Dates (from/to): _____

Terms of employment:
(pay/ hours of work) _____

Duties and responsibilities: _____

I have explained to company about the accredited Internship programme.
Yes / No

I have attached the job description.
Yes / No

18. Appendix H – Weekly Activity Report

Practicum/ Research Practicum/ Internship Project - Weekly Activity Report

Student Name: _____ Student ID: _____

Week Commencing: _____

The weekly activity report is a half a page report that provides a summary of the research activities performed by the student as part of the research project and what the student has learned during the week. The weekly report must be presented and discussed with the supervisor during the timetabled supervision sessions. Supervisor Feedback must also be included. The student is responsible to upload the report on the module Moodle page at the end of the supervision meeting.

Summary of research activities (to be completed by the student)

Supervisor Feedback (to be completed by the student)

19. Appendix I – Student Survey – Pre-Internship

(A softcopy version of this survey will be sent to you prior to Internship commencement)

Student Survey – Pre-Internship

Strongly agree [5]
Agree [4]
Neutral [3]
Disagree [2]
Strongly Disagree [1]

It was easy to find my Internship	[5]	[4]	[3]	[2]	[1]
I am looking forward to my Internship	[5]	[4]	[3]	[2]	[1]
I feel confident that I am well prepared for my Internship	[5]	[4]	[3]	[2]	[1]
I feel satisfied with the support given by NCI to help me prepare for my Internship	[5]	[4]	[3]	[2]	[1]
I have a clear understanding of the duties/tasks I will be expected to complete as part of my Internship	[5]	[4]	[3]	[2]	[1]
I feel confident in my abilities to undertake the duties and fulfill the requirements of my Internship	[5]	[4]	[3]	[2]	[1]
I feel satisfied with the length of my Internship	[5]	[4]	[3]	[2]	[1]

Comments

20. Appendix J – Student Self-Assessment Evaluation

Student's Surname:

Forename(s):

Study Area:

Student I.D. No:

Company name:

Ratings explained:

5	Excellent	Always demonstrates this ability/consistently exceeds expectations
4	Good	Usually demonstrates this ability/sometimes exceeds expectations
3	Fair	Sometimes demonstrates this ability/ meets expectations
2	Poor	Seldom demonstrates this ability/rarely meets expectations
1	Unsatisfactory	Never demonstrates this ability/does not meet expectations

JOB DESCRIPTION: Assessment of performance in aspects of job as given in Internship Agreement.

Please rate your performance in each area by entering a number (1-5) in each box:

If you have not performed any of the duties listed, indicate using N/A (not applicable).

DUTY	Rating
Other: not specified in job description but undertaken whilst on Internship (please indicate):	

Employability Skills:

Looking back on your work experiences and learning from them is important in building up a greater self-awareness of the strengths and weaknesses in your skills-set. This assessment will assist you later in successfully projecting yourself at interviews. Please indicate in the form below the standard to which you feel you have developed each skill during your Internship

Please rate your employability skills by entering a number (1-5) in each box:

1 Ability to Learn

a	Asks pertinent and purposeful questions	[5]	[4]	[3]	[2]	[1]
b	Seeks out opportunities and utilises appropriate resources	[5]	[4]	[3]	[2]	[1]
c	Accepts responsibility for mistakes and learns from experiences	[5]	[4]	[3]	[2]	[1]
d	Seeks feedback on personal development from colleagues	[5]	[4]	[3]	[2]	[1]

2 Reading/Writing/Communication Skills

a	Reads/Comprehends/Follows written materials	[5]	[4]	[3]	[2]	[1]
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b	Communicates ideas and concepts clearly in writing	[5]	[4]	[3]	[2]	[1]
c	Works within procedures appropriate to the job	[5]	[4]	[3]	[2]	[1]
3 Listening and Oral Communications Skills						
a	Listens to others in an effective and attentive manner	[5]	[4]	[3]	[2]	[1]
b	Effectively participates in meetings and/or group settings	[5]	[4]	[3]	[2]	[1]
c	Demonstrates effective verbal communication skills	[5]	[4]	[3]	[2]	[1]
4 Creative Thinking and Problem-Solving Skills						
a	Breaks down complex tasks/problems into manageable pieces	[5]	[4]	[3]	[2]	[1]
b	Brainstorms/develops options and ideas	[5]	[4]	[3]	[2]	[1]
c	Demonstrates an analytical capacity	[5]	[4]	[3]	[2]	[1]
<u>Professional Skills:</u>						
5 Professional and Career Development Skills						
a	Exhibits a self-motivated approach to work	[5]	[4]	[3]	[2]	[1]
b	Demonstrates ability to set appropriate priorities/goals	[5]	[4]	[3]	[2]	[1]
c	Exhibits professional behaviour and attitude	[5]	[4]	[3]	[2]	[1]
6 Interpersonal and Teamwork skills						
a	Manages and resolves conflict in an effective manner	[5]	[4]	[3]	[2]	[1]
b	Supports and contributes to a team atmosphere	[5]	[4]	[3]	[2]	[1]
c	Demonstrates assertive but appropriate behaviour	[5]	[4]	[3]	[2]	[1]
<u>Organisational Skills</u>						
7 Organisational Effectiveness Skills						
a	Seeks to understand and support the organisations mission/goals	[5]	[4]	[3]	[2]	[1]
b	Fits in with the norms and expectations in the organisation	[5]	[4]	[3]	[2]	[1]
c	Works within appropriate decision-making channels	[5]	[4]	[3]	[2]	[1]
d	Is sensitive to organisational politics	[5]	[4]	[3]	[2]	[1]
8 Basic Work Habits						
a	Reports to work as scheduled and on time	[5]	[4]	[3]	[2]	[1]
b	Exhibits a positive and constructive attitude	[5]	[4]	[3]	[2]	[1]
c	Dress and appearance are appropriate for this organisation	[5]	[4]	[3]	[2]	[1]
d	Is clear about what the organisation expects	[5]	[4]	[3]	[2]	[1]
9 Cultural Adaptation (International Internships only)						
a	Is conscious of different cultural values	[5]	[4]	[3]	[2]	[1]
b	Is sensitive to others with different cultural backgrounds	[5]	[4]	[3]	[2]	[1]
c	Uses the working environment to extend his level of cultural awareness	[5]	[4]	[3]	[2]	[1]

ACADEMIC REQUIREMENTS:

Please indicate your status with respect to each of the following elements:

Completion of work-based learning elements	[5]	[4]	[3]	[2]	[1]
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GENERAL FEEDBACK/COMMENTS:

Is the work challenging?	[5]	[4]	[3]	[2]	[1]
Is there appropriate support/supervision?	[5]	[4]	[3]	[2]	[1]
Are there well-defined work targets to be achieved/projects to be completed	[5]	[4]	[3]	[2]	[1]
Career Goals – is the Internship helping to clarify career objectives	[5]	[4]	[3]	[2]	[1]

- Would you consider that this has been a successful Internship, both from a Company and Student point of view? – Explain in terms of how you have both benefited from the Internship

- How do you feel this Internship will benefit you in the future?

- Do you have any suggestions for other student Internships of this kind in the future?

Student's Signature:

Date:

Title/Position:

Telephone: